

Archive: Mean Queue Lengths and Mean Waiting Times (removed from the main thesis; preserved verbatim)

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This file collects the mean-queue-length and mean-waiting-time derivations for the five solved models, removed from the main thesis to keep the focus on the generating-function structure. They are reproduced here unchanged. Cross-references to the main document do not resolve in this standalone file.

1 Model-A (baseline, $\gamma_1 = \gamma_2 = 0, \theta_1 = \theta_2 = 0$)

1.1 Mean queue lengths and mean waiting times

The results for Model-A are well-studied and standard. However, they will here appear in a slightly different way compared to what usually appears on textbooks —e.g. Adan & Resing [?], since our states are defined with the number of customers in the *queues* with a busy server, as opposed to those in the system. The customer in service thus contributes to all those with a factor of ρ to the standard results we would typically find.

1.1.1 Class-1 marginal PGF and mean queue length

Setting $y = 1$ in the fundamental equation (??) gives

$$[-\lambda_1(x-1)(x-1/\rho_1)] P(x, 1) = \mu(x-1) P_y(1).$$

Cancelling the common factor $(x - 1)$ for $x \neq 1$ and rearranging:

$$P(x, 1) = \frac{\mu P_y(1)}{\mu - \lambda_1 x}.$$

The constant $P_y(1)$ is obtained using normalisation for all busy states $P(1, 1) = 1 - \pi_0 = \rho$:

$$\rho = \frac{\mu P_y(1)}{\mu - \lambda_1} \implies P_y(1) = \rho(1 - \rho_1),$$

so that

$$P(x, 1) = \frac{\rho(1 - \rho_1)}{1 - \rho_1 x}.$$

Differentiating by the quotient rule and evaluating at $x = 1$:

$$\mathbb{E}[N_1] = \left. \frac{d}{dx} P(x, 1) \right|_{x=1} = \left. \frac{\rho(1 - \rho_1) \cdot \rho_1}{(1 - \rho_1 x)^2} \right|_{x=1} = \frac{\rho(1 - \rho_1) \rho_1}{(1 - \rho_1)^2} = \frac{\rho \rho_1}{1 - \rho_1}. \quad (1.1)$$

1.1.2 Total queue length from the diagonal

Setting $x = y = z$ in the kernel polynomial $\lambda_1 x^2 - (\lambda_1 + \lambda_2 + \mu)x + \mu$ shows that $x^*(z) = z$ iff $z \in \{1, 1/\rho\}$; hence $x^*(z) \neq z$ for all $z \in (0, 1)$ with $\rho < 1$. Therefore the ratio $(z - x^*(z))/(x^*(z) - z) = -1$ throughout $(0, 1)$, and the formula of Theorem ?? reduces to:

$$P(z, z) = \frac{z(1 - z) \rho(1 - \rho)}{D(z, z)} \cdot (-1).$$

The denominator at $x = y = z$ is $D(z, z) = (1 + \rho)z^2 - z - \rho z^3 = -\rho z(z - 1)(z - 1/\rho)$, so:

$$\begin{aligned} P(z, z) &= \frac{-z(1 - z) \rho(1 - \rho)}{-\rho z(z - 1)(z - 1/\rho)} \\ &= \frac{(1 - z)(1 - \rho)}{(z - 1)(z - 1/\rho)} = \frac{-(1 - \rho)}{z - 1/\rho} = \frac{\rho(1 - \rho)}{1 - \rho z}, \end{aligned} \quad (1.2)$$

confirming the diagonal identity of Corollary ?. Since $\left. \frac{d}{dz} P(z, z) \right|_{z=1} = \frac{\partial}{\partial x} P(1, 1) + \frac{\partial}{\partial y} P(1, 1) = \mathbb{E}[N_1] + \mathbb{E}[N_2]$, differentiating by the quotient rule and evaluating at $z = 1$ gives:

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] = \left. \frac{d}{dz} \frac{\rho(1 - \rho)}{1 - \rho z} \right|_{z=1} = \left. \frac{\rho(1 - \rho) \cdot \rho}{(1 - \rho z)^2} \right|_{z=1} = \frac{\rho^2(1 - \rho)}{(1 - \rho)^2} = \frac{\rho^2}{1 - \rho}.$$

This is the $M/M/1(\lambda, \mu)$ queue-length formula for the combined system [?], as expected: the two classes together behave as a single $M/M/1$ queue.

1.1.3 Class-2 mean queue length

By subtracting the marginal for class-1 to the total number of customers, we find the marginal for class-2:

$$\begin{aligned} \mathbb{E}[N_2] &= (\mathbb{E}[N_1] + \mathbb{E}[N_2]) - \mathbb{E}[N_1] = \frac{\rho^2}{1 - \rho} - \frac{\rho \rho_1}{1 - \rho_1} \\ &= \frac{\rho^2(1 - \rho_1) - \rho \rho_1(1 - \rho)}{(1 - \rho)(1 - \rho_1)} = \frac{\rho[\rho(1 - \rho_1) - \rho_1(1 - \rho)]}{(1 - \rho)(1 - \rho_1)} \\ &= \frac{\rho(\rho - \rho_1)}{(1 - \rho)(1 - \rho_1)} = \frac{\rho \rho_2}{(1 - \rho_1)(1 - \rho)}. \end{aligned} \quad (1.3)$$

1.1.4 Mean waiting times via Little's Law

To obtain the expected waiting times, we can simply apply *Little's Law* ?? to each class independently: we then have $\mathbb{E}[N_i] = \lambda_i \mathbb{E}[W_i]$ where $\mathbb{E}[W_i]$ is the mean waiting time in the queue, excluding the time in service. Therefore:

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1} = \frac{\rho \rho_1}{\lambda_1(1 - \rho_1)} = \frac{\rho \cdot (\lambda_1/\mu)}{\lambda_1(1 - \rho_1)} = \frac{\rho}{\mu(1 - \rho_1)}, \quad (1.4)$$

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2} = \frac{\rho \rho_2}{\lambda_2(1-\rho_1)(1-\rho)} = \frac{\rho \cdot (\lambda_2/\mu)}{\lambda_2(1-\rho_1)(1-\rho)} = \frac{\rho}{\mu(1-\rho_1)(1-\rho)}. \quad (1.5)$$

As expected, class-1 benefits from the priority discipline, and $\mathbb{E}[W_2]$ carries the additional penalty factor $1/(1-\rho)$ compared to $\mathbb{E}[W_1]$.

2 Model- B_2 (one-way jockeying, $\gamma_1 > 0$, $\gamma_2 = 0$)

2.1 Mean queue lengths and mean waiting times

Total queue length from the diagonal

Setting $x = y = z$ in the reduced fundamental equation (??), the jockeying convection term $\gamma_1 xy(x-y)\partial_x P$ vanishes at $x = y = z$ (factor $(x-y) = 0$), as does the boundary term $\mu(x-y)P_y(y)$. The remaining balance is identical to Model- A 's, so the same diagonal cancellation gives

$$P(z, z) = \frac{\rho(1-\rho)}{1-\rho z},$$

the same formula as (1.2). Since $P(z, z) = \mathbb{E}[z^{N_1+N_2}] = \mathbb{E}[z^N]$ is the PGF of the total in-queue count N , the generating-function identity $\frac{d}{dz}P(z, z)|_{z=1} = \mathbb{E}[N_1] + \mathbb{E}[N_2]$ holds; differentiating by the quotient rule and evaluating at $z = 1$:

$$\begin{aligned} \mathbb{E}[N_1] + \mathbb{E}[N_2] &= \frac{d}{dz} \frac{\rho(1-\rho)}{1-\rho z} \Big|_{z=1} = \frac{\rho(1-\rho)}{(1-\rho z)^2} \rho \Big|_{z=1} \\ &= \frac{\rho^2(1-\rho)}{(1-\rho)^2} = \frac{\rho^2}{1-\rho}. \end{aligned}$$

One-way jockeying does not alter the total mean queue length relative to Model- A , since the diagonal PGF is unchanged.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x}|_{(1,1)}$ by first reducing the fundamental equation (??) at $y = 1$, and then differentiating the resulting marginal PGF at $x = 1$.

Step 1: marginal PGF $P(x, 1)$. Evaluating (??) at $y = 1$, the bracket multiplying P becomes $(\lambda_1 + \mu)x - \mu - \lambda_1 x^2 = -\lambda_1(x-1)(x-1/\rho_1)$, the convection coefficient is $\gamma_1 x(x-1)$, and the right-hand side reduces to $\mu(x-1)P_y(1)$ (the $\pi(0, 0)$ term carries the factor $(1-y) = 0$). Every term therefore shares the common factor $(x-1)$; cancelling it leaves the first-order linear ODE in x :

$$-\lambda_1(x-1/\rho_1)P(x, 1) + \gamma_1 x \frac{d}{dx}P(x, 1) = \mu P_y(1).$$

The integrating factor for this ODE is $e^{-\lambda_1 x/\gamma_1} x^{\mu/\gamma_1}$ (noting $\lambda_1/\rho_1 = \mu$), and integrating from 0 to x (the boundary term vanishes since $0^{\mu/\gamma_1} = 0$):

$$P(x, 1) e^{-\lambda_1 x/\gamma_1} x^{\mu/\gamma_1} = \frac{\mu P_y(1)}{\gamma_1} \int_0^x e^{-\lambda_1 t/\gamma_1} t^{\mu/\gamma_1-1} dt.$$

Rearranging and writing $\alpha = \mu/\gamma_1$ and $\mathcal{H}(x) = \int_0^x e^{-\lambda_1 t/\gamma_1} t^{\alpha-1} dt$:

$$P(x, 1) = \frac{\mu P_y(1)}{\gamma_1} e^{\lambda_1 x/\gamma_1} x^{-\alpha} \mathcal{H}(x).$$

The constant $P_y(1)$ follows from $P(1, 1) = 1 - \pi_0 = \rho$, which gives $P_y(1) = \rho \gamma_1 / (\mu e^{\lambda_1/\gamma_1} \mathcal{H}(1))$.

Step 2: differentiate $P(x, 1)$ at $x = 1$. Writing $P(x, 1) = C \cdot f(x) \cdot \mathcal{H}(x)$ with $C = \mu P_y(1)/\gamma_1$ and $f(x) = e^{\lambda_1 x/\gamma_1} x^{-\alpha}$, the product rule together with Leibniz's rule ($d\mathcal{H}/dx = e^{-\lambda_1 x/\gamma_1} x^{\alpha-1}$, the integrand at the upper limit) gives

$$\frac{d}{dx}P(x, 1) = C \left[f'(x) \mathcal{H}(x) + f(x) e^{-\lambda_1 x/\gamma_1} x^{\alpha-1} \right].$$

The logarithmic derivative $f'(x)/f(x) = \lambda_1/\gamma_1 - \alpha/x = (\lambda_1 x - \mu)/(\gamma_1 x)$, and the second product term simplifies as $f(x) e^{-\lambda_1 x/\gamma_1} x^{\alpha-1} = x^{-1}$, so

$$\frac{d}{dx} P(x, 1) = \frac{\mu P_y(1)}{\gamma_1} \left[e^{\lambda_1 x/\gamma_1} x^{-\alpha} \frac{\lambda_1 x - \mu}{\gamma_1 x} \mathcal{H}(x) + x^{-1} \right].$$

Step 3: evaluate at $x = 1$.

$$\begin{aligned} \mathbb{E}[N_1] &= \left. \frac{d}{dx} P(x, 1) \right|_{x=1} \\ &= \frac{\mu P_y(1)}{\gamma_1} \left[e^{\lambda_1/\gamma_1} \frac{\lambda_1 - \mu}{\gamma_1} \mathcal{H}(1) + 1 \right], \end{aligned} \quad (2.1)$$

where $\mathcal{H}(1) = \int_0^1 e^{-\lambda_1 t/\gamma_1} t^{\mu/\gamma_1-1} dt = (\gamma_1/\lambda_1)^{\mu/\gamma_1} \gamma(\mu/\gamma_1, \lambda_1/\gamma_1)$ and $\gamma(\cdot, \cdot)$ is the lower incomplete gamma function. The factor $(\lambda_1 - \mu) < 0$ (since $\rho_1 = \lambda_1/\mu < 1$), confirming that $e^{\lambda_1/\gamma_1}(\lambda_1 - \mu)\mathcal{H}(1)/\gamma_1 + 1 > 0$. No elementary simplification of (2.1) is known; the expression is evaluated numerically. As $\gamma_1 \rightarrow 0^+$ the Model-A value (1.1) is recovered.

Class-2 mean queue length by subtraction

$$\mathbb{E}[N_2] = (\mathbb{E}[N_1] + \mathbb{E}[N_2]) - \mathbb{E}[N_1] = \frac{\rho^2}{1 - \rho} - \mathbb{E}[N_1].$$

Since jockeying defections reduce the class-1 queue, $\mathbb{E}[N_1]$ is smaller than the Model-A value $\rho\rho_1/(1 - \rho_1)$, so $\mathbb{E}[N_2]$ correspondingly exceeds the Model-A value.

Mean waiting times via Little's Law

Little's Law applies here without qualification. It is a rate-conservation identity — for any subsystem in which customers are neither created nor destroyed internally, the time-average number present equals the long-run entry rate times the mean time an entrant spends inside — and it holds for *any* positive-recurrent system with finite mean queue length (here ensured by $\rho < 1$, which gives $\mathbb{E}[N_1] \leq \mathbb{E}[N] = \rho^2/(1 - \rho) < \infty$ and so makes the jockeying rate $\gamma_1 \mathbb{E}[N_1]$ below a genuine finite rate), requiring neither Poisson input nor independence. In particular it remains valid even though one-way jockeying has destroyed the per-class Poisson/renewal structure (the very obstruction that ruled out a probabilistic route in Step (ii)). We apply it to each queue (the waiting line, excluding the customer in service), with $\mathbb{E}[W_i]$ understood as the mean time a customer spends in Queue i averaged over all customers that enter it — an arrival routed directly into service contributing a zero-length visit.

Class 1. Every class-1 arrival is counted as an entrant to Queue 1 at rate λ_1 (those finding the server idle enter service at once, contributing zero waiting time), and each leaves Queue 1 either by entering service or by jockeying to Queue 2. Since no other stream feeds Queue 1, its throughput is exactly λ_1 , and Little's Law gives

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1}. \quad (2.2)$$

Class 2. Queue 2 has *two* input streams: the class-2 Poisson arrivals at rate λ_2 and the jockeying transfers from Queue 1. Because each of the N_1 waiting class-1 customers jockeys independently at rate γ_1 , the long-run transfer rate is $\gamma_1 \mathbb{E}[N_1]$ (a genuine rate by the ergodic theorem, equal to the time average of the instantaneous jockeying intensity $\gamma_1 N_1$). The throughput of Queue 2 is therefore $\lambda_2 + \gamma_1 \mathbb{E}[N_1]$, and Little's Law reads $\mathbb{E}[N_2] = (\lambda_2 + \gamma_1 \mathbb{E}[N_1]) \mathbb{E}[W_2]$, so that

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2 + \gamma_1 \mathbb{E}[N_1]}. \quad (2.3)$$

Here $\mathbb{E}[W_2]$ is the mean time in the class-2 line averaged over *all* Queue-2 entrants of both streams; a jockeyed customer's Queue-2 clock starts at the transfer instant, so its total wait is an initial Queue-1 segment (of mean $\mathbb{E}[W_1]$) followed by this Queue-2 segment. Thus (2.3) is the per-Queue-2-entrant residence time, not the end-to-end delay of a tagged class-2 *arrival*.

The jockeying inflow is what distinguishes (2.3) from the single-stream Models A and C_2 : the naive denominator λ_2 would undercount the entrants to Queue 2 and is correct only in the no-jockeying limit. Indeed, as $\gamma_1 \rightarrow 0^+$ the transfer rate $\gamma_1 \mathbb{E}[N_1] \rightarrow 0$, so (2.3) reduces to $\mathbb{E}[N_2]/\lambda_2$ and both (2.2) and (2.3) recover the Model-A values (1.4)–(1.5).

3 Model- C_2 (class-1 abandonment, $\theta_1 > 0$)

3.1 Mean queue lengths and mean waiting times

Why the diagonal no longer simplifies

Setting $x = y = z$ in (??), the derivative term of abandonment $\theta_1 xy(x-1) \partial_x P$ survives (the factor $(x-1)|_{x=z} \neq 0$ for $z \in (0, 1)$), and the diagonal equation reads

$$[(\lambda_1 + \lambda_2 + \mu)z^2 - \mu z - \lambda z^3]P(z, z) + \theta_1 z^2(z-1) \frac{\partial P}{\partial x}(z, z) = -\mu z(1-z) \pi(0, 0). \quad (3.1)$$

The contrast with Model- A —or any model without abandonment—is instructive: $P(z, z)$ is no longer purely algebraic with closed form $P(z, z) = \pi(0, 0)/(1 - \rho z)$ (1.2), from which the total $\mathbb{E}[N_1] + \mathbb{E}[N_2]$ followed by a single differentiation. Here the surviving derivative term remains as a second unknown, so that route is no longer available; abandonment removes customers and lowers both means relative to models without it.

Class-2 mean queue length via $P(1, y)$

Setting $x = 1$ in (??) we obtain

$$\lambda_2 y(1-y) P(1, y) = \mu(1-y) [P_y(y) - \pi(0, 0)].$$

Cancelling $(1-y)$ for $y \neq 1$ and substituting the known $P_y(y)$ (??),

$$P(1, y) = \frac{\mu [P_y(y) - \pi(0, 0)]}{\lambda_2 y} = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{1 - \tilde{B}_C(\lambda_2(1-y))}{\tilde{B}_C(\lambda_2(1-y)) - y},$$

the marginal PGF of class-2. It is 0/0 at $y = 1$, so set $\varepsilon = 1 - y \rightarrow 0$ and use the LST Taylor expansion $\tilde{B}_C(\lambda_2 \varepsilon) = 1 - \lambda_2 \mathbb{E}[B_C] \varepsilon + \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2 + O(\varepsilon^3)$:

$$\begin{aligned} 1 - \tilde{B}_C(\lambda_2 \varepsilon) &= \lambda_2 \mathbb{E}[B_C] \varepsilon - \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2 + O(\varepsilon^3), \\ \tilde{B}_C(\lambda_2 \varepsilon) - y &= (1 - \lambda_2 \mathbb{E}[B_C] \varepsilon + \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2) - (1 - \varepsilon) = (1 - \lambda_2 \mathbb{E}[B_C]) \varepsilon + \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2 + O(\varepsilon^3). \end{aligned}$$

Writing $a = \lambda_2 \mathbb{E}[B_C]$ and $c = \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2]$,

$$P(1, 1 - \varepsilon) = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{a \varepsilon - c \varepsilon^2 + O(\varepsilon^3)}{(1 - a) \varepsilon + c \varepsilon^2 + O(\varepsilon^3)} = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{a - c \varepsilon}{(1 - a) + c \varepsilon} + O(\varepsilon^2).$$

Differentiating $f(\varepsilon) = (a - c \varepsilon)/((1 - a) + c \varepsilon)$ by the quotient rule,

$$f'(\varepsilon) = \frac{-c((1 - a) + c \varepsilon) - (a - c \varepsilon)c}{((1 - a) + c \varepsilon)^2} = \frac{-c}{(1 - a + c \varepsilon)^2},$$

and since $\frac{d}{dy} P(1, y)|_{y=1} = -\frac{d}{d\varepsilon} P(1, 1 - \varepsilon)|_{\varepsilon=0}$,

$$\mathbb{E}[N_2] = \frac{d}{dy} P(1, y) \Big|_{y=1} = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{c}{(1 - a)^2} = \frac{\mu \pi(0, 0) \lambda_2 \mathbb{E}[B_C^2]}{2(1 - \lambda_2 \mathbb{E}[B_C])^2}. \quad (3.2)$$

As $\theta_1 \rightarrow 0^+$, $\mathbb{E}[B_C] \rightarrow (\mu - \lambda_1)^{-1}$ and $\mathbb{E}[B_C^2] \rightarrow 2\mu/(\mu - \lambda_1)^3$, recovering the Model- A value $\mathbb{E}[N_2] = \rho \rho_2 / ((1 - \rho_1)(1 - \rho))$.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x} \Big|_{(1,1)}$ by evaluating the theorem formula at $y = 1$ and then differentiating in x .

Step 1: marginal PGF $P(x, 1)$. As $y \rightarrow 1$, the exponent $\beta(y) \rightarrow 0$, so $(1 - x)^{\beta(y)} \rightarrow 1$ for fixed $x \in (0, 1)$, and $\mathcal{H}_1(x, y) \rightarrow \int_0^x e^{-\lambda_1 t / \theta_1} t^{\mu / \theta_1 - 1} dt$ while $\mathcal{H}_2(x, y)$ tends to a finite limit. In the theorem's bracket the two integrals carry different weights: $\tilde{B}_C(\lambda_2(1 - y)) \rightarrow 1$, while $1 - \tilde{B}_C(\lambda_2(1 - y)) = \lambda_2 \mathbb{E}[B_C] (1 - y) + O((1 - y)^2) \rightarrow 0$, so the $\mathcal{H}_2$ term drops and the bracket

collapses to $\mathcal{H}_1(x, 1)$. The prefactor's y -dependent part satisfies $(1 - y)/(\tilde{B}_C(\lambda_2(1 - y)) - y) \rightarrow 1/(1 - \lambda_2\mathbb{E}[B_C])$, so

$$P(x, 1) = \frac{\mu \pi(0, 0)}{\theta_1 (1 - \lambda_2\mathbb{E}[B_C])} e^{\lambda_1 x / \theta_1} x^{-\mu / \theta_1} \int_0^x e^{-\lambda_1 t / \theta_1} t^{\mu / \theta_1 - 1} dt.$$

Step 2: differentiate $P(x, 1)$ at $x = 1$. Write $P(x, 1) = C f(x) \mathcal{K}(x)$ with $C = \mu \pi(0, 0)/(\theta_1 (1 - \lambda_2\mathbb{E}[B_C]))$, $f(x) = e^{\lambda_1 x / \theta_1} x^{-\mu / \theta_1}$ and $\mathcal{K}(x) = \int_0^x e^{-\lambda_1 t / \theta_1} t^{\mu / \theta_1 - 1} dt$. By the product rule with Leibniz's rule ($\mathcal{K}'(x) = e^{-\lambda_1 x / \theta_1} x^{\mu / \theta_1 - 1}$) and the logarithmic derivative $f'(x)/f(x) = (\lambda_1 x - \mu)/(\theta_1 x)$,

$$\frac{d}{dx} P(x, 1) = C \left[f'(x) \mathcal{K}(x) + f(x) e^{-\lambda_1 x / \theta_1} x^{\mu / \theta_1 - 1} \right] = C \left[e^{\lambda_1 x / \theta_1} x^{-\mu / \theta_1} \frac{\lambda_1 x - \mu}{\theta_1 x} \mathcal{K}(x) + x^{-1} \right].$$

Step 3: evaluate at $x = 1$.

$$\mathbb{E}[N_1] = \left. \frac{d}{dx} P(x, 1) \right|_{x=1} = \frac{\mu \pi(0, 0)}{\theta_1 (1 - \lambda_2\mathbb{E}[B_C])} \left[e^{\lambda_1 / \theta_1} \frac{\lambda_1 - \mu}{\theta_1} \mathcal{K}(1) + 1 \right],$$

where $\mathcal{K}(1) = \int_0^1 e^{-\lambda_1 t / \theta_1} t^{\mu / \theta_1 - 1} dt = (\theta_1 / \lambda_1)^{\mu / \theta_1} \gamma(\mu / \theta_1, \lambda_1 / \theta_1)$, with $\gamma(\cdot, \cdot)$ the lower incomplete gamma function. Since $\lambda_1 - \mu < 0$ ($\rho_1 < 1$) the bracket is positive. No elementary simplification is known, so $\mathbb{E}[N_1]$ is evaluated numerically; as $\theta_1 \rightarrow 0^+$ it recovers the Model-A value (1.1).

Mean waiting times via Little's Law

The relation $\mathbb{E}[N_i] = \lambda_i \mathbb{E}[W_i]$ holds for any stable system. Here λ_i is the arrival rate of *all* class- i customers, including those that eventually abandon, so $\mathbb{E}[W_i]$ is the mean queueing time averaged over both served and abandoned customers:

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1}, \quad \mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2}.$$

Using the throughput $\mu(1 - \pi_0)$ instead would give the conditional waiting time of served customers only—a different object. As $\theta_1 \rightarrow 0^+$ both means recover the Model-A values (1.1)–(1.5).

4 Model- B_2^H (head-of-line jockeying)

4.1 Mean queue lengths and mean waiting times

4.1.1 Class-1 marginal PGF and mean queue length

Setting $y = 1$ in the fundamental equation (??), the term $\mu x(1 - y)\pi(0, 0)$ vanishes and the coefficient of $P(x, 1)$ factors as $-(x - 1)(\lambda_1 x - (\mu + \gamma_1))$, while the right-hand side becomes $(x - 1)(\mu + \gamma_1)P_y(1)$. Cancelling $(x - 1)$ for $x \neq 1$,

$$P(x, 1) = \frac{(\mu + \gamma_1) P_y(1)}{\mu + \gamma_1 - \lambda_1 x} = \frac{P_y(1)}{1 - \rho_B x}, \quad \rho_B := \frac{\lambda_1}{\mu + \gamma_1}. \quad (4.1)$$

This is a geometric class-1 marginal: the priority queue behaves as an $M/M/1$ queue whose effective *clearance rate* is $\mu + \gamma_1$, since a waiting class-1 customer at the head of Queue 1 departs that queue either by entering service (rate μ) or by jockeying (rate γ_1). The constant $P_y(1)$ is pinned by the normalisation $P(1, 1) = 1 - \pi_0 = \rho$, giving $P_y(1) = \rho(1 - \rho_B)$, so that $P(x, 1) = \rho(1 - \rho_B)/(1 - \rho_B x)$. Differentiating by the quotient rule and evaluating at $x = 1$,

$$\mathbb{E}[N_1] = \left. \frac{d}{dx} P(x, 1) \right|_{x=1} = \frac{\rho(1 - \rho_B) \rho_B}{(1 - \rho_B)^2} = \frac{\rho \rho_B}{1 - \rho_B}. \quad (4.2)$$

This is Model-A's formula (1.1) with the priority load ρ_1 replaced by the *jockeying-reduced* load $\rho_B = \lambda_1/(\mu + \gamma_1) \leq \rho_1$.

4.1.2 Total queue length from the diagonal

Because jockeying conserves the total number of customers, the combined process $N_1 + N_2$ (together with the in-service customer) is the queue-length process of an ordinary $M/M/1(\lambda, \mu)$ queue, and the diagonal generating function (??) is $P(z, z) = \rho(1 - \rho)/(1 - \rho z)$. Since $\frac{d}{dz}P(z, z)|_{z=1} = \mathbb{E}[N_1] + \mathbb{E}[N_2]$, differentiating gives

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] = \frac{d}{dz} \frac{\rho(1 - \rho)}{1 - \rho z} \Big|_{z=1} = \frac{\rho^2(1 - \rho)}{(1 - \rho)^2} = \frac{\rho^2}{1 - \rho}, \quad (4.3)$$

identical to the combined $M/M/1$ result of Model-A (1.3).

4.1.3 Class-2 mean queue length by subtraction

Subtracting (4.2) from (4.3),

$$\mathbb{E}[N_2] = \frac{\rho^2}{1 - \rho} - \frac{\rho \rho_B}{1 - \rho_B} = \frac{\rho(\rho - \rho_B)}{(1 - \rho)(1 - \rho_B)}. \quad (4.4)$$

The excess load admits a transparent decomposition,

$$\rho - \rho_B = \frac{\lambda_1 + \lambda_2}{\mu} - \frac{\lambda_1}{\mu + \gamma_1} = \frac{(\lambda_1 + \lambda_2)(\mu + \gamma_1) - \lambda_1 \mu}{\mu(\mu + \gamma_1)} = \frac{\lambda_2 \mu + \lambda_1 \gamma_1 + \lambda_2 \gamma_1}{\mu(\mu + \gamma_1)} = \rho_2 + \frac{\lambda_1 \gamma_1}{\mu(\mu + \gamma_1)},$$

so $\rho - \rho_B$ exceeds the bare class-2 load ρ_2 precisely by the load that jockeying transfers from Queue 1 into Queue 2. At $\gamma_1 = 0$ it collapses to ρ_2 and (4.4) recovers Model-A's $\mathbb{E}[N_2] = \rho \rho_2 / [(1 - \rho_1)(1 - \rho)]$.

4.1.4 Mean waiting times via Little's Law

We apply **Little's Law** to each queue separately, recalling that N_i counts only the customers *waiting*, so $\mathbb{E}[N_i] = \Lambda_i \mathbb{E}[W_i]$ with Λ_i the long-run rate at which customers *enter* Queue i and $\mathbb{E}[W_i]$ their mean residence time in that queue (from entry until they leave it, by service entry or by jockeying).

Every class-1 customer enters Queue 1 exactly once, by exogenous arrival, so $\Lambda_1 = \lambda_1$ and

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1} = \frac{\rho \rho_B}{\lambda_1(1 - \rho_B)} = \frac{\rho}{\mu + \gamma_1 - \lambda_1}.$$

Compared with Model-A's $\mathbb{E}[W_1] = \rho/(\mu - \lambda_1)$ (1.4), the denominator is enlarged by γ_1 : head-of-line jockeying drains the priority queue faster and shortens class-1 waits.

Queue 2 receives a *mixed* input: the exogenous Poisson stream at rate λ_2 together with the jockeying stream. The latter fires at rate γ_1 whenever $N_1 \geq 1$, so its long-run rate is $\gamma_1 \mathbb{P}(N_1 \geq 1)$. The probability that a class-1 customer is waiting follows from the geometric marginal (4.1) by subtracting the $N_1 = 0$ mass from the busy mass,

$$\mathbb{P}(N_1 \geq 1) = P(1, 1) - P(0, 1) = \rho - \rho(1 - \rho_B) = \rho \rho_B,$$

since $P(1, 1) = 1 - \pi_0 = \rho$ and $P(0, 1) = P_y(1) = \rho(1 - \rho_B)$. The total entrance rate into Queue 2 is therefore $\Lambda_2 = \lambda_2 + \gamma_1 \rho \rho_B$, and Little's Law gives the mean residence time per Queue-2 entrant,

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\Lambda_2} = \frac{\rho(\rho - \rho_B)}{(1 - \rho)(1 - \rho_B)(\lambda_2 + \gamma_1 \rho \rho_B)}. \quad (4.5)$$

As $\gamma_1 \rightarrow 0^+$ the jockeying stream disappears, $\Lambda_2 \rightarrow \lambda_2$ and $\rho_B \rightarrow \rho_1$, and (4.5) recovers Model-A's $\mathbb{E}[W_2] = \rho/[\mu(1 - \rho_1)(1 - \rho)]$ (1.5).

5 Model- C_2^H (head-of-line abandonment)

5.1 Mean queue lengths and mean waiting times

5.1.1 Class-1 marginal PGF and mean queue length

The class-1 marginal is the geometric law (??); pinning the constant by $P(1, 1) = 1 - \pi_0$ as in (??) gives $P(x, 1) = (1 - \pi_0)(1 - \rho_C)/(1 - \rho_C x)$. Differentiating and evaluating at $x = 1$,

$$\mathbb{E}[N_1] = \frac{d}{dx} P(x, 1) \Big|_{x=1} = \frac{(1 - \pi_0) \rho_C}{1 - \rho_C}, \quad \rho_C = \frac{\lambda_1}{\mu + \theta_1}, \quad (5.1)$$

with $1 - \pi_0 = \lambda(\mu + \theta_1)/[\mu(\mu + \theta_1) + \lambda_1\theta_1]$ from Corollary ?? . As in Model- B_2^H , the priority queue behaves as an $M/M/1$ queue with effective clearance rate $\mu + \theta_1$, the only difference from Model- B_2^H 's (4.2) being the prefactor $1 - \pi_0$, which here is no longer equal to ρ because abandonment alters the idle probability.

5.1.2 Class-2 mean queue length

Abandonment is a true departure and so breaks the *conservation of the total customer count* that underlies the diagonal argument: in Model- B_2^H the jockeying factor $\gamma_1 y(x - y)$ vanishes on the diagonal $x = y$, whereas here the abandonment factor $\theta_1 y(x - 1)$ does not. Concretely, setting $x = y = z$ in (??) leaves the term $\theta_1 z(z - 1)P_y(z)$ on the right-hand side, so the diagonal equation no longer closes and the route that furnished $\mathbb{E}[N_1] + \mathbb{E}[N_2]$ for Model- B_2^H is unavailable. We instead differentiate the class-2 marginal. From the marginal relation (??), $P(1, y) = \mu[P_y(y) - \pi(0, 0)]/(\lambda_2 y)$, and since $\mathbb{E}[N_2] = \frac{d}{dy}P(1, y)|_{y=1}$,

$$\mathbb{E}[N_2] = \frac{\mu}{\lambda_2} P'_y(1) - (1 - \pi_0), \quad (5.2)$$

where we used $\frac{\mu}{\lambda_2}[P_y(1) - \pi(0, 0)] = P(1, 1) = 1 - \pi_0$. It remains to compute $P'_y(1)$ from (??). Both the numerator $\mu x^*(y)(1 - y)\pi(0, 0)$ and the denominator $\Delta(y) = (\mu + \theta_1 y)x^*(y) - y(\mu + \theta_1)$ vanish at $y = 1$ (since $x^*(1) = 1$ gives $\Delta(1) = (\mu + \theta_1) - (\mu + \theta_1) = 0$), so $P_y(y)$ is a 0/0 form there. Writing $N(y)$ and $\Delta(y)$ for numerator and denominator, a second-order Taylor expansion about $y = 1$ gives the value and the slope

$$P_y(1) = \frac{N'(1)}{\Delta'(1)} = (1 - \pi_0)(1 - \rho_C), \quad P'_y(1) = \frac{N''(1)\Delta'(1) - N'(1)\Delta''(1)}{2\Delta'(1)^2},$$

which is why both the first and the second root sensitivities of x^* at $y = 1$ are needed, namely (??) together with

$$\frac{d^2}{dy^2}x^*(1) = \frac{2(\mu + \theta_1)\lambda_2^2}{(\mu + \theta_1 - \lambda_1)^3}$$

(the latter obtained by differentiating $x^*(y)$ in (??) twice). Substituting these into $P'_y(1)$, then into (5.2), and simplifying yields the closed form

$$\mathbb{E}[N_2] = \frac{\lambda_2 \lambda (\mu + \theta_1) [(\mu + \theta_1)^2 - \lambda_1 \theta_1]}{(\mu + \theta_1 - \lambda_1) [\mu(\mu + \theta_1) + \lambda_1 \theta_1] [(\mu + \theta_1)(\mu - \lambda) + \lambda_1 \theta_1]}, \quad (5.3)$$

with $\lambda = \lambda_1 + \lambda_2$. The last bracket in the denominator is exactly the stability numerator of π_0 from (??), so $\mathbb{E}[N_2]$ is finite precisely under the stability condition (??) and diverges as $\pi_0 \rightarrow 0^+$. As $\theta_1 \rightarrow 0^+$ the right-hand side reduces to $\lambda_2 \lambda / [(\mu - \lambda_1)(\mu - \lambda)] = \rho \rho_2 / [(1 - \rho_1)(1 - \rho)]$, recovering Model- A 's (1.3).

5.1.3 Mean waiting times via Little's Law

We apply **Little's Law** to each queue. Every class-1 customer enters Queue 1 once by exogenous arrival and later leaves it either by entering service or by abandoning from the head; in both cases it leaves Queue 1, so the entrance rate is $\Lambda_1 = \lambda_1$ and

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1} = \frac{(1 - \pi_0)\rho_C}{\lambda_1(1 - \rho_C)}.$$

Here $\mathbb{E}[W_1]$ is the mean time a class-1 customer spends *waiting* in Queue 1, averaged over all class-1 arrivals, whether they are eventually served or abandon. Queue 2, by contrast, receives no jockeying input and its customers never abandon ($\theta_2 = 0$), so every class-2 arrival is eventually served and the entrance rate is simply $\Lambda_2 = \lambda_2$:

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2}$$

with $\mathbb{E}[N_2]$ from (5.3), the λ_2 cancelling against its numerator. Both means reduce to the corresponding Model- A expressions as $\theta_1 \rightarrow 0^+$.